Lecture 9

Root systems, Weyl groups and weight lattices

Objectives. *The roots of a semisimple Lie algebra satisfy a few axioms about reflections. We study these axioms abstractly. A choice of positive roots gives simple roots, a fundamental chamber and an order on weights. Simple reflections move every vector into the closure of that chamber.*

By the end of this lecture you will be able to:

- Prove that the simple roots form a basis in which every root has integer coordinates of one sign.
- Prove that the simple reflections generate the Weyl group.
- Compute Q, P, the fundamental weights and the dominant weight in a Weyl orbit.
- Prove that only finitely many dominant weights lie below a dominant weight.

Throughout, E is a finite-dimensional real vector space with an inner product $(\,\cdot\,,\cdot\,)$. For nonzero $\alpha \in E$, define the *coroot* $\alpha^\vee$ and the pairing with it by

$$\alpha^\vee = \frac{2\alpha}{(\alpha,\alpha)}, \qquad \langle x, \alpha^\vee \rangle = (x, \alpha^\vee) = \frac{2(x,\alpha)}{(\alpha,\alpha)}.$$

9.1 Root systems and reflections

The *reflection* in a nonzero vector α is the linear map $s_\alpha : E \to E$ given by

$$s_\alpha(x) = x - \langle x, \alpha^\vee \rangle \alpha.$$

The reflection s_α fixes the hyperplane $H_\alpha = \{x \in E : (x,\alpha) = 0\}$ and sends α to $-\alpha$. Since $E = H_\alpha \oplus \mathbb{R}\alpha$ is an orthogonal decomposition, s_α is orthogonal and $s_\alpha^2 = 1$.

Definition 9.1 (Root system)**.** A *root system* in E is a finite subset $R \subset E$ such that

(R1) $0 \notin R$, and R spans E;

(R2) $\mathbb{R}\alpha \cap R = \{\alpha, -\alpha\}$ for every $\alpha \in R$;

(R3) $s_\alpha(R) = R$ for every $\alpha \in R$;

(R4) $\langle \beta, \alpha^\vee \rangle \in \mathbb{Z}$ for all $\alpha, \beta \in R$.

The *Weyl group* W is the subgroup of $\mathrm{GL}(E)$ generated by the reflections s_α with $\alpha \in R$.

Let $\mathfrak{g}$ be semisimple with Cartan subalgebra $\mathfrak{h}$. Lecture 8 proved that its roots form a root system in their real span $E \subset \mathfrak{h}^*$, with the form induced by the Killing form. A positive

multiple of this form gives the same reflections and pairings. For these roots, $\langle \beta, \alpha^\vee \rangle = \beta(h_\alpha)$, and (R3) is the root-string relation.

Lecture 8 defined the coroot of α as the element h_α of $\mathfrak{h}$. Here $\alpha^\vee$ is a vector of E. The Killing form connects the two. It identifies $\mathfrak{h}^*$ with $\mathfrak{h}$ by the map $\lambda \mapsto t_\lambda$ of that lecture, and for the Killing-induced form this map sends $2\alpha/(\alpha,\alpha)$ to h_α. The two coroots therefore correspond, and we use the symbol $\alpha^\vee$ for both. A positive multiple of the form changes the vector in E and the identification, but not the element h_α.

The group W is finite. By (R3), each $w \in W$ permutes R. An element fixing every root fixes a spanning set, so it is the identity. Thus W embeds in the permutation group of R.

Each $w \in W$ is orthogonal. Let $\alpha \in E$ be nonzero and $x \in E$. Expanding $s_\alpha(w^{-1}x)$ gives the first line below, and the orthogonality of w gives the second:

$$\begin{aligned} w s_\alpha w^{-1}(x) &= x - \langle w^{-1}x, \alpha^\vee \rangle \, w\alpha \\ &= x - \langle x, (w\alpha)^\vee \rangle \, w\alpha \\ &= s_{w\alpha}(x). \end{aligned} \tag{9.1}$$

Lemma 9.2 (Acute roots · *[P]*). *Let $\alpha, \beta \in R$ with $\beta \neq \pm\alpha$. If $(\alpha,\beta) > 0$, then $\alpha - \beta \in R$.*

Proof. By (R2), α and β are not proportional. The Cauchy–Schwarz inequality is therefore strict:

$$\langle \alpha, \beta^\vee \rangle \langle \beta, \alpha^\vee \rangle = \frac{4(\alpha,\beta)^2}{(\alpha,\alpha)(\beta,\beta)} < 4.$$

Both factors are positive, since $(\alpha,\beta) > 0$, and they are integers by (R4). Their product is less than 4, so one of them equals 1.

If $\langle \alpha, \beta^\vee \rangle = 1$, then $s_\beta(\alpha) = \alpha - \beta \in R$. If $\langle \beta, \alpha^\vee \rangle = 1$, then $s_\alpha(\beta) = \beta - \alpha \in R$. In that case $\alpha - \beta \in R$ too, because $R = -R$ by (R3). ∎

9.2 Positive and simple roots

The finitely many hyperplanes H_α do not cover E, since a finite union of proper subspaces of a real vector space has empty interior. Choose $\gamma \in E$ with $(\gamma,\alpha) \neq 0$ for every $\alpha \in R$.

Definition 9.3 (Positive and simple roots). The *positive roots* for γ are

$$R_+ = \{\alpha \in R : (\gamma,\alpha) > 0\}.$$

A positive root is *simple* if it is not the sum of two positive roots. Write $\Delta = \{\alpha_1, \dots, \alpha_n\}$ for the set of simple roots and $s_i = s_{\alpha_i}$ for the *simple reflections*.

Every root is positive or negative, so $R = R_+ \sqcup (-R_+)$. For $\mathfrak{g}$, the choice of R_+ is the choice of $\mathfrak{n}_+ = \bigoplus_{\alpha \in R_+} \mathfrak{g}_\alpha$.

Theorem 9.4 (Simple roots form a basis · *[P]*). *(a) Every $\alpha \in R_+$ has the form $\alpha = \sum_i k_i \alpha_i$ with all $k_i \in \mathbb{Z}_{\geq 0}$.*

(b) If $i \neq j$, then $(\alpha_i, \alpha_j) \leq 0$.

(c) The set Δ is a basis of E.

Proof. (a) Suppose the claim fails. Choose a counterexample α with (γ,α) minimal. It is not simple, so $\alpha = \beta + \beta'$ with $\beta, \beta' \in R_+$. Since $(\gamma,\alpha) = (\gamma,\beta) + (\gamma,\beta')$ with both terms positive,

$$0 < (\gamma,\beta) < (\gamma,\alpha) \quad \text{and} \quad 0 < (\gamma,\beta') < (\gamma,\alpha).$$

By minimality, β and β' are nonnegative integer combinations of Δ. So is α, a contradiction.

(b) Suppose $(\alpha_i, \alpha_j) > 0$. Lemma 9.2 gives $\alpha_i - \alpha_j \in R$. If this root is positive, then $\alpha_i = (\alpha_i - \alpha_j) + \alpha_j$ is not simple. If it is negative, then $\alpha_j = (\alpha_j - \alpha_i) + \alpha_i$ is not simple. Both cases are impossible.

(c) By (a), the span of Δ contains R, hence equals E. For independence, suppose $\sum_i c_i\alpha_i = 0$. Let $I = \{i : c_i > 0\}$ and $J = \{j : c_j < 0\}$, and put

$$v = \sum_{i\in I} c_i\alpha_i = \sum_{j\in J}(-c_j)\alpha_j.$$

The sets I and J are disjoint. By (b),

$$(v, v) = \sum_{i\in I}\sum_{j\in J} c_i(-c_j)(\alpha_i, \alpha_j) \le 0.$$

Hence $v = 0$. Then $0 = (\gamma, v) = \sum_{i\in I} c_i(\gamma, \alpha_i)$, a sum of positive terms. So $I = \varnothing$. Similarly $J = \varnothing$. ∎

Since Δ is a basis, the coordinates in (a) are unique. A negative root is the negative of a positive root. Hence every root has all coordinates ≥ 0 or all ≤ 0. The *height* of $\alpha = \sum_i k_i\alpha_i$ is $\operatorname{ht}\alpha = \sum_i k_i$.

Lemma 9.5 (Simple reflections and ρ · *[P]*)**.** *The reflection s_i permutes $R_+ \setminus \{\alpha_i\}$. Consequently, for $\rho = \frac{1}{2}\sum_{\alpha\in R_+}\alpha$,*

$$s_i\rho = \rho - \alpha_i, \qquad \langle\rho, \alpha_i^\vee\rangle = 1.$$

Proof. Let $\beta \in R_+$ with $\beta \ne \alpha_i$. By (R2), β is not a multiple of α_i. So $\beta = \sum_l k_l\alpha_l$ has some $k_j > 0$ with $j \ne i$. The root

$$s_i\beta = \beta - \langle\beta, \alpha_i^\vee\rangle\alpha_i$$

has the same coefficient $k_j > 0$. The coordinates of a root have one sign, so $s_i\beta$ is positive, and $s_i\beta \ne \alpha_i$. Since $s_i^2 = 1$, the map s_i is a bijection of $R_+ \setminus \{\alpha_i\}$.

Hence s_i fixes the sum of the roots in $R_+ \setminus \{\alpha_i\}$ and sends α_i to $-\alpha_i$. Thus

$$s_i(2\rho) = 2\rho - 2\alpha_i, \qquad \text{so} \qquad s_i\rho = \rho - \alpha_i.$$

The definition of s_i gives $s_i\rho = \rho - \langle\rho, \alpha_i^\vee\rangle\alpha_i$. Comparing the coefficients of α_i gives $\langle\rho, \alpha_i^\vee\rangle = 1$. ∎

9.3 Weyl chambers and generation

Definition 9.6 (Weyl chambers)**.** A point of E is *regular* if it lies on no hyperplane H_α. A *Weyl chamber* is a connected component of the set of regular points. The *fundamental chamber* is

$$C = \{x \in E : (x, \alpha_i) > 0 \text{ for } i = 1, \dots, n\}.$$

The *walls* of C are the hyperplanes $H_{\alpha_1}, \dots, H_{\alpha_n}$. A vector x is *dominant* if $(x, \alpha_i) \ge 0$ for every i.

Figure 9.1 previews these terms for the root system A_2, which § 9.5 works out. The proofs below do not use the figure. Here $\gamma = \alpha_1 + \alpha_2$. The roots at an acute angle to γ are the positive roots, and the lines are the hyperplanes H_α.

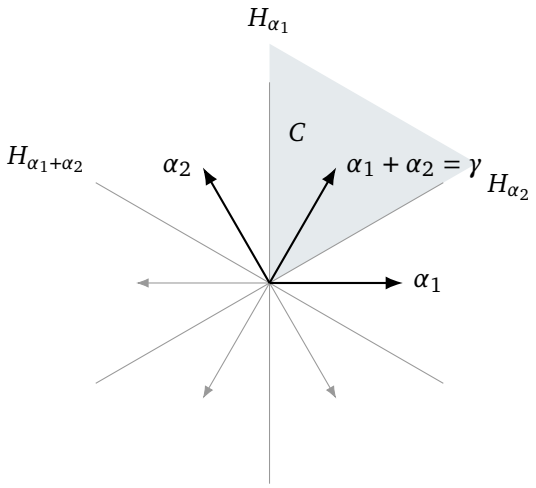


Figure 9.1: A preview of A_2. Dark arrows are the positive roots for γ, and light arrows are the negative roots. The shaded sector is part of the fundamental chamber C, whose walls are H_{α_1} and H_{α_2}.

Proposition 9.7 (The fundamental chamber · *[P]*)**.** *The set C is a Weyl chamber. Its closure $\overline{C}$ is the set of dominant vectors.*

Proof. By the positive-root expansion in Theorem 9.4, a point of C is positive on every positive root. So C consists of regular points. The set C contains ρ, because $(\rho, \alpha_i) = \frac{1}{2}(\alpha_i, \alpha_i) > 0$ by Lemma 9.5. The set C is convex, hence connected.

Let D be the chamber containing C. For each $\alpha \in R$, the function $x \mapsto (x, \alpha)$ is continuous and nonzero on the connected set D. So this function has constant sign on D, namely its sign at ρ. Hence $D \subseteq C$, and $D = C$.

The dominant vectors form a closed set containing C. Hence $\overline{C}$ consists of dominant vectors. Conversely, if y is dominant, then $y + t\rho \in C$ for $t > 0$, and $y + t\rho \to y$ as $t \to 0$. So $y \in \overline{C}$. ∎

Let $W_0 \subseteq W$ be the subgroup generated by $s_1, \ldots, s_n$.

Proposition 9.8 (Dominant conjugates · *[P]*)**.** *For every $x \in E$ there is $w \in W_0$ with $wx \in \overline{C}$. Consequently every Weyl chamber has the form $w^{-1}C$ with $w \in W_0$.*

Proof. The group W_0 is finite. Choose $w \in W_0$ with (wx, ρ) maximal. For each i, maximality applied to $s_i w \in W_0$, the orthogonality of s_i and Lemma 9.5 give

$$(wx, \rho) \geq (s_i wx, \rho) = (wx, s_i \rho) = (wx, \rho) - (wx, \alpha_i).$$

Hence $(wx, \alpha_i) \geq 0$ for every i.

Each $w \in W$ permutes the hyperplanes H_α, since $w(H_\alpha) = H_{w\alpha}$. So w maps chambers homeomorphically onto chambers. Let D be a chamber and $x \in D$. Choose $w \in W_0$ with $wx \in \overline{C}$. The point wx is regular, so $wx \in C$. The chamber wD contains wx, so $wD = C$ and $D = w^{-1}C$. ∎

Theorem 9.9 (Generation by simple reflections · *[P]*)**.** *Every root is W_0-conjugate to a simple root, and $W = W_0$.*

Proof. Let $\beta \in R_+$. Among the positive roots in $W_0\beta$, choose $\beta' = \sum_i k_i\alpha_i$ of minimal height. Then

$$0 < (\beta', \beta') = \sum_i k_i(\beta', \alpha_i),$$

so $(\beta', \alpha_i) > 0$ for some i, because all $k_i \geq 0$. Suppose $\beta' \neq \alpha_i$. Then $s_i\beta' \in W_0\beta$ is positive by Lemma 9.5. Since $\langle \beta', \alpha_i^\vee \rangle > 0$,

$$\operatorname{ht}(s_i\beta') = \operatorname{ht}\beta' - \langle \beta', \alpha_i^\vee \rangle < \operatorname{ht}\beta'.$$

This contradicts minimality. Hence $\beta' = \alpha_i$.

A negative root β has $-\beta = w\alpha_i$ with $w \in W_0$. Then $\beta = ws_i\alpha_i$.

Finally, let $\beta \in R$ and write $\beta = w\alpha_i$ with $w \in W_0$. Equation (9.1) gives $s_\beta = ws_iw^{-1} \in W_0$. So every generator s_β of W lies in W_0, and $W = W_0$. ∎

It remains to show that W acts *freely* on the chambers. In other words, only $w = 1$ maps a chamber to itself. The proof uses reduced words and a length function, so we cite it.

Theorem 9.10 (Chamber theorem · *[C]*)**.** *Let R satisfy (R1)–(R4), and let Δ be its simple roots for R_+.*

(a) If $w \in W$ and $w\Delta = \Delta$, then $w = 1$.

(b) If $\lambda, \mu \in \overline{C}$ and $w\lambda = \mu$ for some $w \in W$, then w is a product of simple reflections fixing λ. In particular, $\lambda = \mu$.

Source: Humphreys, Section 10.3, Theorem (e) and Lemma B. Humphreys calls a subset $\Delta \subseteq R$ a *base* if it is a basis of E in which every root has integer coordinates of one sign. Theorem 9.4 shows that our Δ is a base.

Let $w \in W$ with $wC = C$. Since R_+ consists of the roots positive on C, we get $wR_+ = R_+$. So w permutes Δ, which is defined from R_+ alone, and the chamber theorem gives $w = 1$. Every chamber has the form vC (Proposition 9.8). If $wvC = vC$, then $v^{-1}wv$ fixes C, so $v^{-1}wv = 1$ and $w = 1$. Hence W acts freely on the chambers. With Proposition 9.8, W acts simply transitively on the chambers.

Bases correspond to chambers. Let Δ' be a base. Since Δ' is a basis of E, we can choose $\gamma' \in E$ with $(\gamma', \alpha) = 1$ for every $\alpha \in \Delta'$. A root has integer coordinates of one sign in Δ', so its pairing with γ' is a nonzero integer of that sign. Hence γ' is regular, and the roots positive on γ' are the nonnegative combinations of Δ'. A sum of two such roots pairs with γ' to at least 2, so each element of Δ' is simple for this choice. The simple roots form a basis by Theorem 9.4, so Δ' is the whole set of simple roots of the chamber containing γ'. An element of W that maps one chamber to another maps the simple roots of the first to those of the second. Hence W acts simply transitively on the bases of R. This result rests on the cited chamber theorem.

9.4 Lattices and dominant weights

Definition 9.11 (Root and weight lattices)**.** The *root lattice* and the *weight lattice* are

$$Q = \bigoplus_{i=1}^{n} \mathbb{Z}\alpha_i, \qquad P = \{\lambda \in E : \langle \lambda, \alpha^\vee \rangle \in \mathbb{Z} \text{ for all } \alpha \in R\}.$$

The *fundamental weights* $\omega_1, \dots, \omega_n \in E$ are defined by

$$\langle \omega_i, \alpha_j^\vee \rangle = \delta_{ij} \quad \text{for } i, j = 1, \dots, n.$$

The *dominant integral weights* form $P_+ = P \cap \overline{C}$. For $\lambda, \mu \in E$, write $\mu \le \lambda$ if

$$\lambda - \mu \in Q_+ = \bigoplus_{i=1}^{n} \mathbb{Z}_{\ge 0}\alpha_i.$$

The ω_i exist and form a basis, since the simple coroots $\alpha_j^\vee$ form a basis. By Theorem 9.4, Q contains R. By (R4), $Q \subseteq P$. For $\mathfrak{g}$, the condition on P reads $\lambda(h_\alpha) \in \mathbb{Z}$.

Proposition 9.12 (Simple coroots suffice · *[P]*)**.** *A vector $\lambda \in E$ lies in P if and only if $\langle \lambda, \alpha_i^\vee \rangle \in \mathbb{Z}$ for every i. Hence*

$$P = \bigoplus_{i=1}^{n} \mathbb{Z}\omega_i, \qquad P_+ = \Big\{ \sum_i m_i\omega_i : m_i \in \mathbb{Z}_{\ge 0} \Big\}, \qquad \alpha_j = \sum_i a_{ij}\omega_i,$$

where the Cartan matrix *has entries $a_{ij} = \langle \alpha_j, \alpha_i^\vee \rangle$.*

Proof. Let $L = \bigoplus_k \mathbb{Z}\alpha_k^\vee$. Since $s_j x = x - (x, \alpha_j)\alpha_j^\vee$, we have

$$s_j(\alpha_k^\vee) = \alpha_k^\vee - \langle \alpha_j, \alpha_k^\vee \rangle \alpha_j^\vee = \alpha_k^\vee - a_{kj}\alpha_j^\vee \in L.$$

Hence $s_j L \subseteq L$ for each j, so $W_0 L = L$. For $\beta \in R$, Theorem 9.9 gives $\beta = w\alpha_i$ with $w \in W_0$. As w is orthogonal, $\beta^\vee = w\alpha_i^\vee \in L$. If λ is integral on every $\alpha_k^\vee$, it is therefore integral on the integer combination $\beta^\vee$ of them. The converse holds because each α_i lies in R.

The three formulas follow from the dual-basis expansion

$$\lambda = \sum_{i=1}^{n} \langle \lambda, \alpha_i^\vee \rangle \omega_i \quad \text{for } \lambda \in E. \qquad \blacksquare$$

By Lemma 9.5, $\rho = \sum_i \omega_i \in P_+$. For $\lambda \in P$ and $\alpha \in R$,

$$s_\alpha \lambda = \lambda - \langle \lambda, \alpha^\vee \rangle \alpha \in \lambda + Q.$$

Since $Q \subseteq P$, the group W preserves P. Moreover, $w\lambda - \lambda \in Q$ for every $w \in W$.

Theorem 9.13 (Dominant representatives · *[P given the chamber theorem]*)**.** *For each $\lambda \in P$, the orbit $W\lambda$ contains exactly one element of P_+.*

Proof. Proposition 9.8 gives w with $w\lambda \in \overline{C}$, and $w\lambda \in P$ because W preserves P. So $W\lambda \cap P_+$ is nonempty. If $\mu, \mu' \in W\lambda \cap P_+$, then $\mu' = w'\mu$ for some $w' \in W$, and Theorem 9.10(b) gives $\mu = \mu'$. $\blacksquare$

The existence half is fully proved here. Lecture 13 needs only this half, through Corollary 9.15 below.

Proposition 9.14 (Finiteness below a dominant weight · *[P]*)**.** *Let $\lambda \in P_+$. The set $\{\mu \in P_+ : \mu \le \lambda\}$ is finite.*

Proof. Let $\mu \in P_+$ with $\lambda - \mu = \sum_i k_i\alpha_i$ and all $k_i \ge 0$. The vector $\lambda + \mu$ is dominant and all $k_i \ge 0$. Hence every term of the last sum below is nonnegative:

$$(\lambda, \lambda) - (\mu, \mu) = (\lambda + \mu, \lambda - \mu) = \sum_i k_i(\lambda + \mu, \alpha_i) \ge 0.$$

So μ lies in the closed ball of radius $|\lambda|$.

The map $T\colon \mathbb{R}^n \to E$ given by $T(m) = \sum_i m_i\omega_i$ is a linear isomorphism. By Proposition 9.12, it maps the integer vectors onto P. Its continuous inverse maps the ball to a bounded set. A bounded set contains only finitely many integer vectors m. ∎

Corollary 9.15 (Weyl-stable sets · *[P]*). *Let $\lambda \in P_+$. Let $S \subseteq P$ be W-stable with $\nu \le \lambda$ for every $\nu \in S$. Then S is finite.*

Proof. For $\nu \in S$, Proposition 9.8 gives a dominant $w\nu \in S$. Then $w\nu \le \lambda$, so $w\nu$ lies in the finite set F of Proposition 9.14. Hence $S \subseteq \bigcup_{\mu\in F} W\mu$, a finite union of finite orbits. ∎

9.5 The root system A_2

Let $E = \{x \in \mathbb{R}^3 : x_1 + x_2 + x_3 = 0\}$ with the standard inner product. Let

$$R = \{e_i - e_j : i \neq j\}, \qquad \gamma = (1, 0, -1).$$

These are the roots of $\mathfrak{sl}_3$ for diagonal $\mathfrak{h}$ (Lecture 7), where $x \in E$ is the linear form $\operatorname{diag}(h_1, h_2, h_3) \mapsto \sum_i x_i h_i$ on $\mathfrak{h}$. On $\mathfrak{sl}_3$ the Killing form is $6\operatorname{tr}(XY)$, so the standard inner product is 6 times the Killing-induced form. Positive rescaling changes no reflection, coroot pairing, chamber or lattice. Since $(\gamma, e_i - e_j) = \gamma_i - \gamma_j$, the positive roots are $e_i - e_j$ with $i < j$. The simple roots are

$$\alpha_1 = e_1 - e_2, \qquad \alpha_2 = e_2 - e_3,$$

and $e_1 - e_3 = \alpha_1 + \alpha_2$. Every root has $(\alpha, \alpha) = 2$, so $\alpha^\vee = \alpha$ and

$$(a_{ij}) = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

The reflection $s_{e_i - e_j}$ exchanges x_i and x_j. Hence $W \cong S_3$ permutes coordinates. Figure 9.2 shows the six chambers cut out by the lines H_α. It also shows part of the weight lattice P and of the root lattice Q, and the orbit $W\omega_1$, which the next examples compute. Each point $a\omega_1 + b\omega_2$ of P is coloured by $a - b$ modulo 3.

Example 9.16 (Two coordinate systems). Since $\alpha_j^\vee = \alpha_j$, the fundamental weights solve $(\omega_i, \alpha_j) = \delta_{ij}$ in E. The solutions are

$$\omega_1 = \frac{1}{3}(2, -1, -1) = \frac{1}{3}(2\alpha_1 + \alpha_2),$$
$$\omega_2 = \frac{1}{3}(1, 1, -2) = \frac{1}{3}(\alpha_1 + 2\alpha_2).$$

The columns of the Cartan matrix give

$$\alpha_1 = 2\omega_1 - \omega_2, \qquad \alpha_2 = -\omega_1 + 2\omega_2.$$

Write $[a, b]$ for $a\omega_1 + b\omega_2$. The positive roots are

$$\alpha_1 = [2, -1], \qquad \alpha_2 = [-1, 2], \qquad \alpha_1 + \alpha_2 = [1, 1] = \rho.$$

The Gram matrix of the fundamental weights and the squared length of $[a, b]$ are

$$\big((\omega_i, \omega_j)\big) = \frac{1}{3}\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \qquad |[a, b]|^2 = \frac{2}{3}(a^2 + ab + b^2).$$

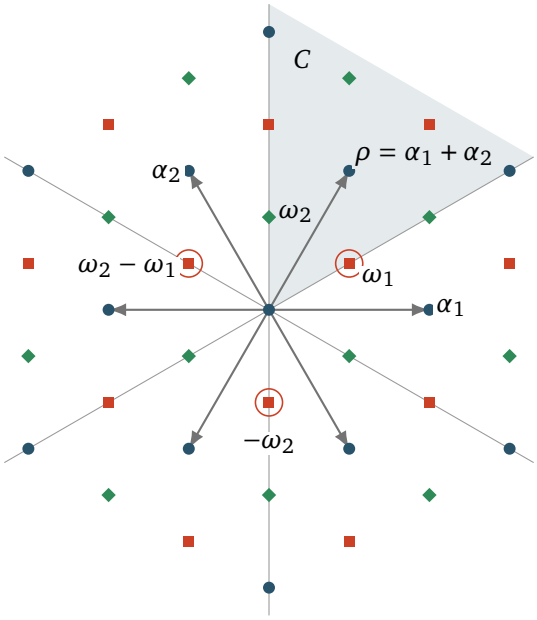


Figure 9.2: The root system A_2 and its lattices. Arrows are the six roots, the three lines are the H_α, and the shaded sector is part of the chamber C. Round dots are the points of Q, squares the class of ω_1 and diamonds the class of ω_2. The rings mark the orbit $W\omega_1$.

Example 9.17 (Q inside P). A weight $[a,b] \in P$ lies in Q if and only if $a \equiv b \pmod 3$. Indeed, if $[a,b] = m\alpha_1 + n\alpha_2$, then

$$a = 2m - n, \qquad b = -m + 2n, \qquad a - b = 3(m-n).$$

Conversely, if $3 \mid a-b$, the integers

$$m = \frac{2a+b}{3} = a - \frac{a-b}{3}, \qquad n = \frac{a+2b}{3} = b + \frac{a-b}{3}$$

solve these equations. Hence $[a,b] \mapsto a - b \bmod 3$ induces $P/Q \cong \mathbb{Z}/3$. The classes are those of 0, ω_1 and ω_2. In particular $\omega_1, \omega_2 \notin Q$.

Example 9.18 (The orbit of ω_1). In these coordinates $\langle [a,b], \alpha_1^\vee \rangle = a$ and $\langle [a,b], \alpha_2^\vee \rangle = b$. The rule $s_i\lambda = \lambda - \langle \lambda, \alpha_i^\vee \rangle \alpha_i$ gives

$$\begin{aligned} s_1[1,0] &= [1,0] - 1\cdot[2,-1] = [-1,1], \\ s_2[-1,1] &= [-1,1] - 1\cdot[-1,2] = [0,-1], \\ s_2[1,0] &= [1,0], \\ s_1[0,-1] &= [0,-1]. \end{aligned}$$

The s_i are involutions, so these three weights form a set stable under s_1 and s_2. By Theorem 9.9 it is the whole orbit:

$$W\omega_1 = \{\omega_1,\ \omega_2 - \omega_1,\ -\omega_2\}.$$

In simple-root coordinates and in $\mathbb{R}^3$,

$$\begin{aligned} \omega_1 &= \frac{1}{3}(2\alpha_1 + \alpha_2) = \frac{1}{3}(2,-1,-1), \\ \omega_2 - \omega_1 &= \frac{1}{3}(-\alpha_1 + \alpha_2) = \frac{1}{3}(-1,2,-1), \\ -\omega_2 &= \frac{1}{3}(-\alpha_1 - 2\alpha_2) = \frac{1}{3}(-1,-1,2). \end{aligned}$$

Only ω_1 is dominant, as Theorem 9.13 requires.

Figure 9.3 shows the finiteness test for $\lambda = 2\omega_1$, which the next example carries out. The circle is the norm bound of Proposition 9.14. The coset $\lambda + Q$ is a discrete set, so the figure marks it point by point.

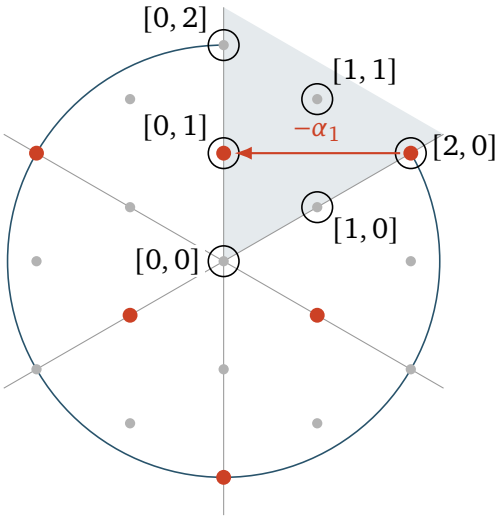


Figure 9.3: The circle has radius $|\lambda|$. The six rings are the dominant weights inside it or on it. Red dots are the coset $\lambda + Q$ inside the circle or on it, and grey dots are the other weights. Only $[2,0]$ and $[0,1] = \lambda - \alpha_1$ are both dominant and in the coset.

Example 9.19 (Dominant weights below $2\omega_1$)**.** Let $\lambda = 2\omega_1 = [2,0]$, so $|\lambda|^2 = \frac{8}{3}$. The proof of Proposition 9.14 bounds a dominant $\mu = [a,b] \le \lambda$ by

$$a^2 + ab + b^2 \le 4, \qquad a, b \ge 0.$$

The solutions are $[0,0]$, $[1,0]$, $[0,1]$, $[1,1]$, $[2,0]$ and $[0,2]$. The condition $\lambda - \mu \in Q$ requires $a - b \equiv 2 \pmod 3$. Only $[0,1]$ and $[2,0]$ remain. Finally,

$$2\omega_1 - \omega_2 = [2,-1] = \alpha_1 \in Q_+.$$

The dominant weights below $2\omega_1$ are $2\omega_1$ and ω_2.

Self-checks

(1) In A_2, write $\omega_2 - \omega_1$ in simple-root coordinates. Is it in Q?
(2) In A_2, find the dominant weight in the orbit of $-\omega_1$.
(3) In A_2, which dominant weights lie below ρ?

Answers. (1) It is $\frac{1}{3}(-\alpha_1 + \alpha_2)$, which is not in Q. Equivalently, $[-1, 1]$ has $a - b = -2 \not\equiv 0 \pmod 3$. (2) $s_1[-1, 0] = [-1, 0] + \alpha_1 = [1, -1]$ and $s_2[1, -1] = [1, -1] + \alpha_2 = [0, 1]$. The answer is ω_2. (3) The bound $a^2 + ab + b^2 \leq 3$ leaves $[0, 0]$, $[1, 0]$, $[0, 1]$, $[1, 1]$. Congruence modulo Q leaves 0 and ρ, and $\rho - 0 = \alpha_1 + \alpha_2 \in Q_+$. The answer is $\{\rho, 0\}$.

Summary

A choice of positive roots gives simple roots. They form a basis in which every root has integer coordinates of one sign. Simple reflections permute the other positive roots, shift ρ by a simple root, and generate W.

A vector lies in the weight lattice exactly when its pairings with the simple coroots are integers. Every Weyl orbit in P meets P_+. By the cited chamber theorem, it meets P_+ exactly once. Only finitely many dominant weights lie below a dominant weight. Lecture 13 uses this finiteness to bound the weights of a highest-weight quotient.

Reading

Humphreys, *Introduction to Lie Algebras and Representation Theory*, Sections 9–10 and 13.1–13.2. Kirillov, *An Introduction to Lie Groups and Lie Algebras*, Chapter 7. Exercise Sheet 9 compares B_2 with C_2 and works through chamber walls and the finiteness lemma.